AgNW Reservoir Computing — Full Manual

Question: **can a silver-nanowire (AgNW) network work as a reservoir computer?** Answer so far: with framework binary junctions or Fig. 4-calibrated filament: **no**; with volatile measured-data junction N10 on 16 pads: **yes, weakly** ($\mathrm{MC} \approx 3.5$, degree-2 ≈ 6.7, NARMA-10 $\mathrm{NMSE} \approx 0.40$), broken by filament ruptures $\geq 0.01\,\mathrm{s}^{-1}$ per junction. Details: `docs/RESULTS.md` (results), `docs/NEXT_STEPS.md` (status, open items), `docs/report_2026-09-29/report.md`, `docs/report_2026-10-06/report.md` and `docs/report_2026-10-07/report.md` (summary reports), `RESUME_FROM_HERE.md` (setup, reading order).

Hard rules:

- Network geometry, wires, Kirchhoff solver and binary junction dynamics come unchanged from `AgNW_FDT_Memristor_Framework` (`agnw_sim`, commit `f6b6951`, installed editable). Import, never copy. Special cases of new code here must reproduce the framework exactly.
- This repo designs electrodes, input encoding, junction variants, readout, tasks, baselines, criteria.
- Run everything with `.venv/bin/python` (tests `.venv/bin/python -m pytest`).
- One BLAS thread per process (`OMP_NUM_THREADS=1`; scripts set it).
- Math in markdown as LaTeX (`scripts/md_math_to_latex.py` fixes leftovers).
- `legacy/` is superseded; its memory-capacity numbers are invalid.

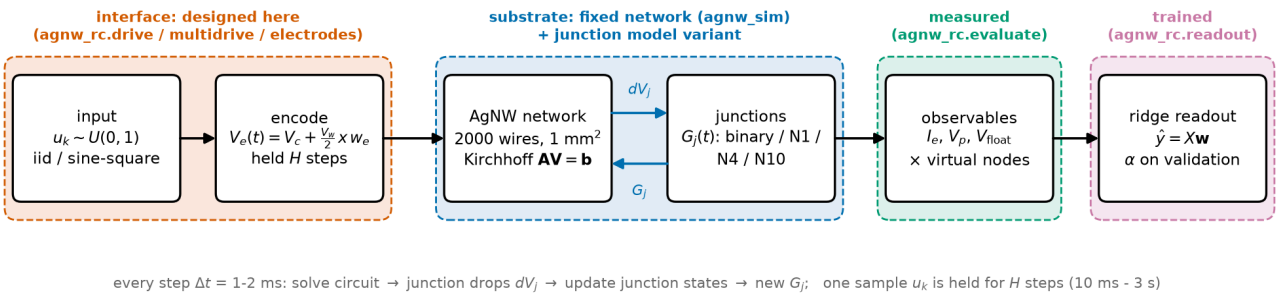


Figure 1. Pipeline. Interface (orange) and readout (pink) are designed or trained here; the network (blue) is the framework's, with a junction model variant (§3). Circuit and junctions are solved in a loop every time step.

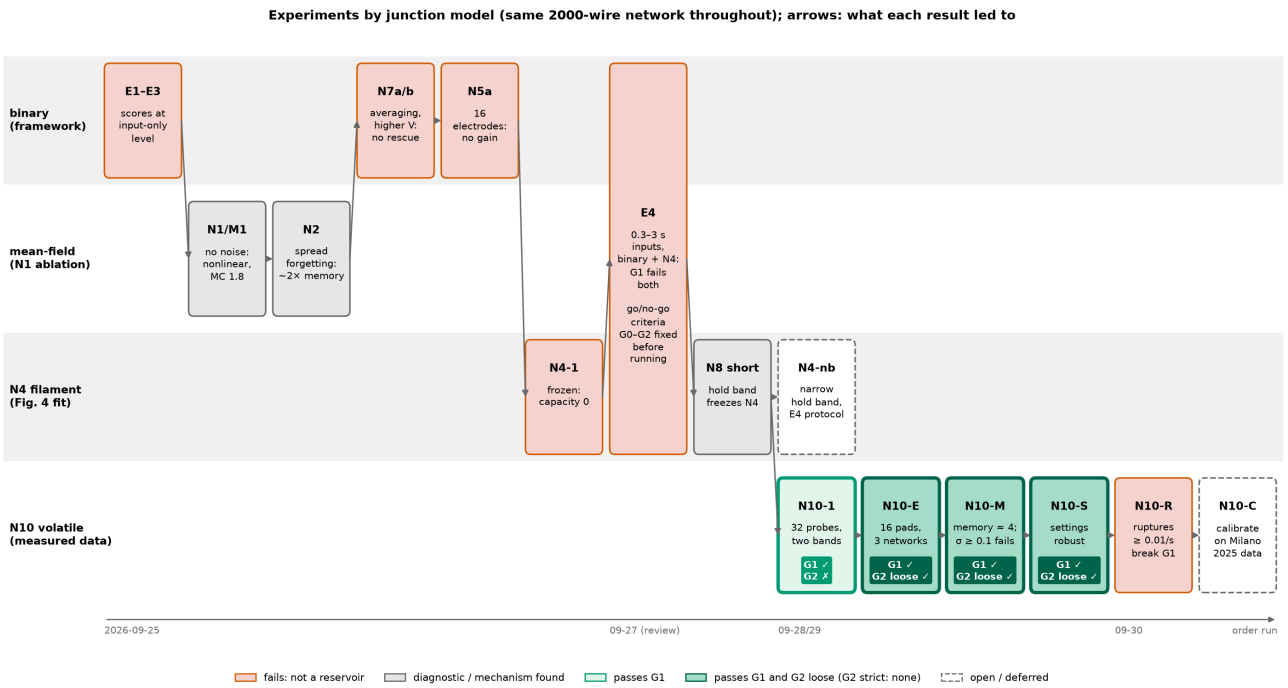


Figure 2. Experiments by junction model in the order they were run, coloured by outcome (details §7.1). Thick green borders and badges mark the experiments that pass G1 (N10-1) and G1 + G2 loose (N10-E/M/S); none passes G2 strict.

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Concepts and glossary

Reservoir computing in one paragraph. A fixed dynamical system (the *reservoir*, here the nanowire network) is driven by an input stream u_k. Its state at each step is a nonlinear function of the recent input history. Only a linear *readout* $\widehat{y} = X\mathbf{w}$ on measured features X is trained, by ridge regression.

The reservoir is useful if (i) the same input history always gives the same state (*echo-state property*, here *consistency*), (ii) the state remembers past inputs (*memory*), and (iii) it mixes them nonlinearly. Each is measured with the standard tasks below and compared with software reservoirs (ESNs) of the same size.

term	meaning
MC	linear memory capacity: $\sum_{k \geq 1} C_k$, how well past inputs $u[t-k]$ can be reconstructed (max = number of features) (§5.3)
IPC2, degree-2	information-processing capacity of degree 2: reconstruction of products/squares of past inputs; nonlinearity measure (§5.2)
NARMA-10	standard benchmark mixing memory (10 steps) and nonlinearity; scored as NMSE (1 = useless, 0 = perfect; input-only ≈ 0.8)
NMSE	mean-square error / target variance; capacity $C = 1 - \mathrm{NMSE}$
ESN	echo-state network: random recurrent $\tanh$ network, the software reference reservoir (§5.4)
consistency	correlation between two runs with the same input but different noise and initial state; 1 = reproducible (§5.3)
G0 / G1 / G2	go/no-go criteria: reproducible / computes beyond its input / useful vs ESN (§5.5)
hold, H	number of 1 ms steps each input sample is held on the electrodes (input duration)
stride	circuit + junction update every `stride` ms ($\Delta t = $ `stride` $\times 1$ ms); hold must be a multiple
washout	drive before scoring so the network forgets its all-OFF start ($\geq 3\times$ the slowest time scale)
virtual nodes	the same current sampled at several times inside one hold; turns transients into extra features (§4.3)
onset	network voltage at which conductance jumps in a slow (quasi-static) sweep: the first conducting paths form
window	input voltage range $[V_c - V_w/2, V_c + V_w/2]$; set relative to the onset
replicate	repeat with the same input and network, different switching noise / initial state; used for consistency
network seed	geometry seed (1, 2, 3): three different random networks

term	meaning
oracle readout	all junction states as features; not measurable, used only to see what information exists inside
probes	32 internal node voltages; would need micro-contacts, reference only
M1	the mean-field (N1) experiment grid (§7.2)
Lamas Fig. 4	I–V loop of the $N = 2000$ network under a 3.6 V sine in Lamas et al. 2026; a *simulation*, used to calibrate N4
E*n* / N*n*	experiment and work-item names (§7.1; E = planned grids, N = follow-ups in `docs/NEXT_STEPS.md`)

Symbols reused with different meanings: r = N4 hold-band ratio $V_{\mathrm{reset}}/V_{\mathrm{set}}$ (rupture rate is r_{rup}); σ = junction noise amplitude (N4, N10) in §3, feature std in §5.1/§5.3, ESN input scale σ_{in} in §5.4.

0. Quickstart

```
.venv/bin/python -m pytest -q                      # 80 tests, ~7 min; checks framework commit too
.venv/bin/python scripts/pilot.py                  # staircase onset + SET/relax steps ->
        results/pilot.npz
.venv/bin/python scripts/run_sims.py e1 --procs 14  # one experiment grid -> results/raw/*.npz
.venv/bin/python scripts/evaluate_all.py           # scores -> results/metrics.json, summary.csv
.venv/bin/python scripts/make_figures.py --only fig3
```

One run in Python:

```
from agnw_rc import config as C
from agnw_rc.multidrive import simulate_multi

wo = C.model_washout_steps("volatile", tau_s=3.0, tau_spread=1.0)   # 66502; grids round up to whole
        holds (67000)
spec = C.RunSpec(stream="iid", hold=1000, v_center=0.38, v_width=0.16, network_seed=1, replicate=0,
                 n_samples=800, model="volatile", layout="pads", tau_s=3.0, tau_spread=1.0,
                 washout=wo, n_virtual=50, stride=2)
raw = simulate_multi(spec)          # dict of arrays, see section 7.3 (hours: 800 x 1 s holds)

from agnw_rc.evaluate import observables, score_iid, noise_floor
F = observables(raw)["electrodes"]  # headline readout of a pad chip (others: "current", "probes",
        "junctions")
s = score_iid(F, raw["u"], floor=noise_floor(raw["u"]))
print(s["MC"], s["IPC2"], s["narma10_nmse"])
```

Criteria (G0–G2) need 3 networks x 2 replicates plus a sine/square run; the report scripts (`n10e_pilot.py report`, `e4_report.py`, ...) do this and write `results/*_criteria.json`.

Read results: `results/summary.csv` (mean/std over nets and replicates) and `results/*_criteria.json` (G0/G1/G2 verdicts, section 5.5).

1. Repository map

```
agnw_rc/     reservoir layer on top of agnw_sim (section 6.1)
scripts/     pilots, parallel runs, reports, checks, figures (section 6.2)
tests/       80 pytest: framework contract, drive, readout, metrics, tasks, baselines,
             mean-field, filament, volatile, multi-electrode, criteria (section 6.3)
docs/        PLAN, RESULTS, NEXT_STEPS, SWITCHING_MECHANISMS, N4_DESIGN, N10_DESIGN,
             REVIEW_2026-09-27, LITERATURE_NOTES, MEASUREMENT_REQUEST, MANUAL (this file),
             report_2026-09-28/, report_2026-09-29/, report_2026-10-06/
memory/      portable project memory (loaded via CLAUDE.md)
legacy/      superseded prototype (its memory-capacity numbers invalid)
figures/     generated plots; figures/manual/ schematics for this manual
results/     raw/*.npz runs (git-ignored, ~6 GB slim; full files with junction states archived, §7.3) +
*.json/csv scores
```

Framework dependency (`../AgNW_FDT_Memristor_Framework/agnw_sim/`): `geometry.py` sticks + intersections, `circuit.py` two-terminal solve, `dualnode.py` dual-node graph + wire resistance, `cached_solver.py` refactor-reuse LU, `dynamics.py` binary SET/RESET, `presets.py` `lamas2026_fig4`, `simulate.py` `run_simulation`, `io.py`, `waveforms.py`, `analysis.py`.

2. Fixed substrate (imported, never copied)

2.1 Geometry — `agnw_sim.geometry`, `agnw_rc.config.make_network`

$N = 2000$ sticks of length $L = 70\,\mu\text{m}$ in a square $A = 1\,\text{mm}^2$ (side $S = \sqrt{A} = 1\,\text{mm}$). Centers $\mathbf{c}_i \sim U([0, S]^2)$, angles $\theta_i \sim U([0, \pi))$, endpoints $\mathbf{p}_{i,1/2} = \mathbf{c}_i \mp (L/2)(\cos\theta_i, \sin\theta_i)$. All pairwise intersections (spatial-hash candidates + segment test) become junctions. Seeded: `build_network(2000, 1e-6, 70e-6, rng(seed))`, seeds $1, 2, 3$. Wire diameter $D = 90\,\text{nm}$ (inferred; reproduces Fig. 4 peak $\approx 8\%$).

2.2 Dual-node circuit — `agnw_sim.circuit`, `agnw_sim.dualnode`

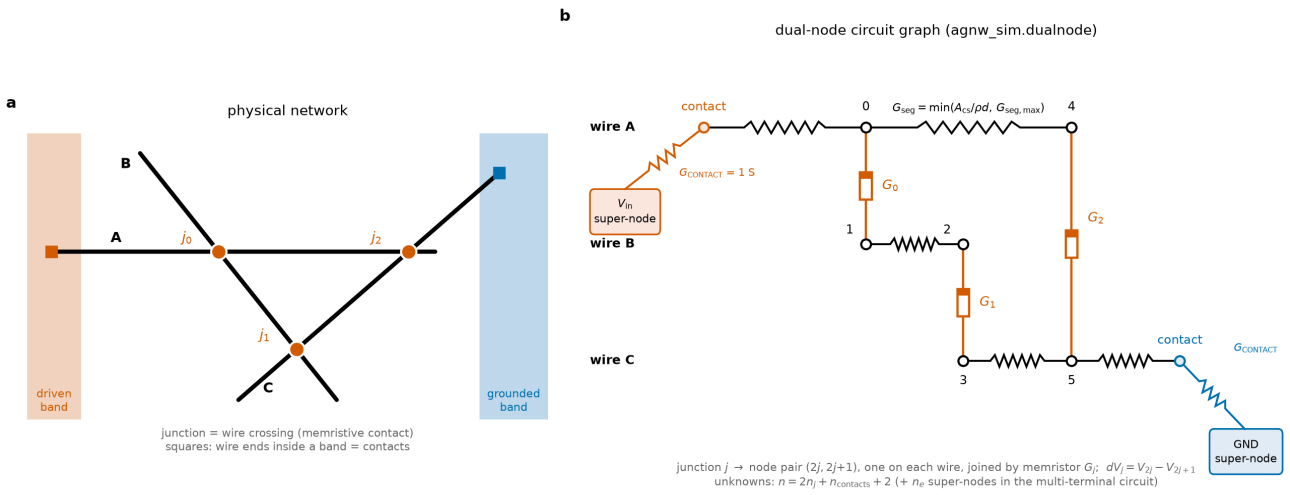


Figure 3. (a) Three wires, three crossings j_0–j_2, contacts where wire ends lie inside a band. (b) The same as a circuit: each junction becomes a node pair on its two wires joined by memristor G_j; wire segments between nodes are resistors G_{seg}; contacts tie to electrode super-nodes through G_{CONTACT}.

Each wire is split at its junctions into resistive segments. Each junction j becomes two graph nodes $(2j, 2j+1)$ on the two wires, joined by the memristor edge G_j. Consecutive nodes on the same wire are joined by a resistive segment with conductance

$$G_{\text{seg}} = \min\left(\frac{A_{\text{cs}}}{\rho\, d},\, G_{\text{seg,max}}\right), \qquad A_{\text{cs}} = \pi(D/2)^2,$$

with Ag resistivity $\rho = 1.59 \times 10^{-8}\,\Omega\text{m}$ (`RHO_AG`), wire diameter $D = 90\,\text{nm}$, distance d along the wire, and cap $G_{\text{seg,max}} = 10^3\,\text{S}$ (`G_SEG_MAX`). Electrode wires get one extra contact node at the wire end inside the band, tied to the electrode by $G_{\text{CONTACT}} = 1.0\,\text{S} \gg G_{\text{on}}$.

Full nodal admittance $\mathbf{Y} \in \mathbb{R}^{n\times n}$, $n = 2n_j + n_{\text{contacts}} + 2$ (last two = V_{in}, GND super-nodes):

$$Y_{pq} = -g_{pq} \quad (p \neq q), \qquad Y_{pp} = \sum_{q\neq p} g_{pq}$$

Dirichlet reduction: fix V_{in}, $V_{\text{gnd}} = 0$, solve free reachable nodes:

$$\mathbf{Y}_{ff}\,\mathbf{V}_f = -\mathbf{Y}_{f,\text{fix}}\,\mathbf{V}_{\text{fix}}$$

Unreachable components (no path to either electrode) set to $0\,\text{V}$. Total current $I = \sum_c G_{\text{CONTACT}}(V_{\text{in}} - V_c)$ over the driven contact nodes c. Framework `DualNodeCircuit.solve()` / `cached_solver` (LU refactor only when G changed). `agnw_rc.config`: `DT = 1e-3` ($\Delta t = 10^{-3}$ s), `P_RELAX = 1e-4` (per step), `SOLVER = "cached"`, `RHO_AG = 1.59e-8`, `G_SEG_MAX = 1e3`.

2.3 Multi-terminal extension — `agnw_rc.multicircuit`

Same graph, but n_e electrode super-nodes (`MultiTerminalCircuit`). $n = 2n_j + n_{\text{contacts}} + n_e$. System $\mathbf{Ax} = \mathbf{b}$ with $\mathbf{A} = \mathbf{L}_{ff} + \mathbf{J}^T\text{diag}(\mathbf{G})\mathbf{J}$, $\mathbf{b} = \mathbf{BV}_{\text{fixed}}$, $\mathbf{B} = -\mathbf{L}_{\text{fixed}}[\text{free}, \text{fixed}]$. `fixed` = driven + grounded super-nodes; floating pads are free unknowns. Methods `reuse` (refactor iff G changed, `splu`), `direct`, `pcg` (`cg` with LU preconditioner, `rtol = 1e-13`, `max_iter = 30`, refactor if `info != 0` or iterations exceed `max_iter / 2`). Tested to match framework matrix to round-off. Junction drops $dV_j = V_{2j} - V_{2j+1}$. Electrode currents, with s_e the super-node of electrode e and $\mathcal{C}_e$ its contact nodes:

$$I_e = \sum_{c\,\in\,\mathcal{C}_e} G_{\text{CONTACT}}\,(V_{s_e} - V_c), \qquad \sum_e I_e = 0.$$

3. Junction models

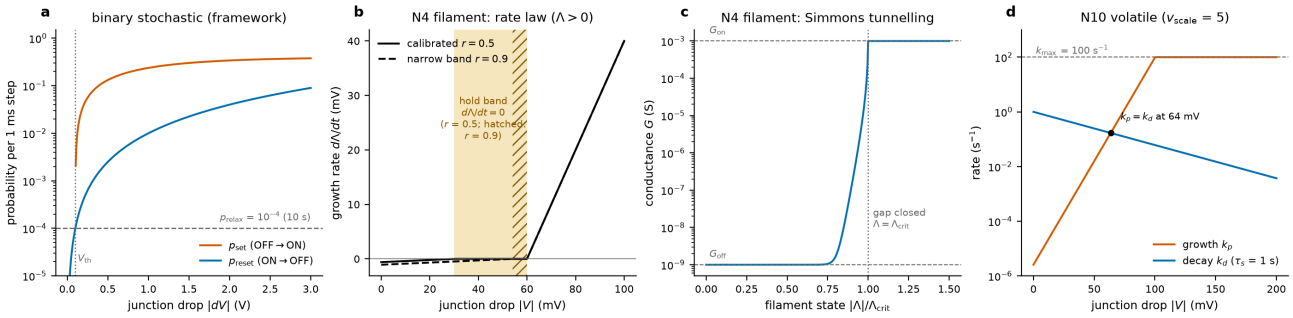


Figure 4. Switching laws, computed with the project code at the parameters used. (a) Binary: per-step SET and RESET probabilities vs junction drop; relaxation 10^{-4} per step. (b) N4: $d\Lambda/dt$ is zero in the hold band $[V_{\text{reset}}, V_{\text{set}}]$ (calibrated $r = 0.5$; hatched: $r = 0.9$), which freezes the network. (c) N4 conductance rises over 6 decades just before the gap closes. (d) N10: growth and decay rates cross at ≈ 64 mV per junction; no hold band.

3.1 Framework binary stochastic (E1–E4, N5a, N7) — `agnw_sim.dynamics`

Verdict: not a reservoir. Switching noise makes junctions irreproducible; forgetting too slow.

State $G_j \in \{G_{\text{off}}, G_{\text{on}}\}$, $G_{\text{off}} = 10^{-9}\,\text{S}$, $G_{\text{on}} = 10^{-3}\,\text{S}$. Lamas 2026 preset (`presets.lamas2026_fig4`, `SimParams.from_step_probs(0.4,0.01)`):

$$P_0 = 400\,\text{s}^{-1}, \quad P_{\text{decay}} = 10\,\text{s}^{-1}\text{unit}^{-2}, \quad \alpha = 1.0\,\text{V}^{-1}, \quad V_{\text{th}} = 0.1\,\text{V}, \quad I_{\text{unit}} = 10^{-3}\,\text{A}$$

($V_{\text{th}} = 0.1\,\text{V}$ inferred from Fig. 4; the paper prints $0.01\,\text{V}$, which moves the network onset to $\approx 0.35\,\text{V}$, far below the Fig. 4 loop.) Per step $\Delta t = 10^{-3}\,\text{s}$:

$$p_{\text{set}}(|dV|) = \Delta t\, P_0 \max\left(0,\, 1 - e^{-\alpha(|dV| - V_{\text{th}})}\right)$$

$$p_{\text{reset}}(I) = \min\left(1,\, \Delta t\, P_{\text{decay}}\, (I/I_{\text{unit}})^2\right), \quad I = G_{\text{on}}\, dV$$

$$p_{\text{off}} = 1 - (1 - p_{\text{reset}})(1 - p_{\text{relax}}), \quad p_{\text{relax}} = 10^{-4}/\text{step}$$

`update_dynamics`: OFF $\rightarrow$ ON w.p. $p_{\text{set}}(|dV|)$; ON $\rightarrow$ OFF w.p. p_{off}. Synchronous from start-of-step state, non-polar ($|dV|$, I^2 only). Mean ON lifetime at zero current 10^4 steps $= 10\,\text{s}$. At E1 drive: ON fraction 0.21–0.32, ≈ 5–7 flips/sample.

Onset values (all $N = 2000$, two bands, different protocols):

protocol	onset
quasi-static staircase from all-OFF (`pilot.py`, RESULTS §1)	$\approx 1.5\,\text{V}$
Fig. 4 sine loop, $3.6\,\text{V}$ (`fig4.py`)	$1.58\,\text{V}$
Lamas Fig. 4 (paper target)	$1.92\,\text{V}$

Why it fails: one noisy slow integrator. Which junction switches is noise-decided (junction consistency 0.00); $10\,\mathrm{s}$ forgetting vs 10–$100\,\mathrm{ms}$ inputs. Scores: $\mathrm{MC} \leq 0.26$ (0.76 at $1\,\mathrm{s}$ holds; 0.94 at $3\,\mathrm{s}$ in E4), degree-2 ≤ 0.13, NARMA ≥ 0.73. Averaging 8 noise replicates (N7a) does not rescue junction/probe features; higher voltages (N7b) do not help.

3.2 Mean-field ablation N1/N2 — `agnw_rc.meanfield`

Verdict: diagnostic only. Removing noise gives reproducibility and nonlinearity, not memory.

State $P_j \in [0,1]$, ON probability:

$$P_j \leftarrow P_j(1-p_{\mathrm{off}}) + (1-P_j)\,p_{\mathrm{set}}$$

exact expectation of §3.1 given voltages, with p_{reset} evaluated at $I_{\mathrm{on}} = G_{\mathrm{on}}|dV|$. Circuit uses $G_j = G_{\mathrm{off}} + P_j(G_{\mathrm{on}} - G_{\mathrm{off}})$. N2 heterogeneous: per-junction $p_{\mathrm{relax},\,j} \sim$ log-uniform $[p_{\mathrm{relax}}, p_{\mathrm{relax,\,max}}]$, quenched, seed $50000+$ network seed. Own PCG solver (`MeanFieldSolver`, `rtol = 1e-13`, `max_iter = 30`) since all G change each step. Onset ≈ 3–$4\,\mathrm{V}$ (no fluctuations to pin at V_{th}; M1 windows 3.5–$5.5\,\mathrm{V}$). Result: reproducible (consistency 1.00), degree-2 2.2, sine/square 1.00, but $\mathrm{MC} \approx 1.8$ ($\rightarrow 2.6$ with spread) vs ESN-32 13.8. Noise blocks nonlinearity; single slow rate blocks memory.

3.3 Continuous filament N4 — `agnw_rc.filament`, `agnw_rc.fig4`

Verdict: reproducible but frozen. The hold band parks almost every junction.

Physical graded junction (Hochstetter 2021 Eqs. 5–7 + 2024 correction). State Λ [V s], $|\Lambda| \leq \Lambda_{\max}$:

$$\frac{d\Lambda}{dt} = \begin{cases} (|V| - V_{\mathrm{set}})\,\mathrm{sgn}(V) & |V| > V_{\mathrm{set}} \\ 0 & V_{\mathrm{reset}} \leq |V| \leq V_{\mathrm{set}} \\ b\,(|V| - V_{\mathrm{reset}})\,\mathrm{sgn}(\Lambda) & |V| < V_{\mathrm{reset}} \end{cases}$$

`unipolar=True`: growth always toward $\Lambda \geq 0$ (survives polarity flip). Gap $s = \max(\Lambda_{\mathrm{crit}} - |\Lambda|, 0)\,s_{\max}/\Lambda_{\mathrm{crit}}$; Simmons low-bias conductance:

$$G_t(s) = A\,\frac{3\sqrt{2m^*}\,e^2\sqrt{e\phi}}{2h^2 s}\exp\left(-4\pi s\sqrt{2m^* e\phi}/h\right)$$

(ϕ in eV value, $m^* = m_{\mathrm{eff}} m_e$; $s = 0 \Rightarrow G_t = \infty$). Total:

$$G(\Lambda) = \frac{1}{R_t + R_f} + G_{\mathrm{off}}, \quad R_t = 1/G_t,\ R_f = 1/G_{\mathrm{on}}$$

rising $G_{\mathrm{off}} \rightarrow G_{\mathrm{on}}$ at $|\Lambda| \geq \Lambda_{\mathrm{crit}}$. Noise (Euler–Maruyama): $\Lambda_{n+1} = \mathrm{clip}(\Lambda_n + \Delta t\, f + \sigma\sqrt{\Delta t}\ \xi_n)$, $\xi \sim \mathcal{N}(0,1)$; deterministic dissolution never crosses 0.

Defaults (`FilamentParams`; code names `lambda_crit`, `lambda_max`): $V_{\mathrm{set}} = 0.01\,\mathrm{V}$, $V_{\mathrm{reset}} = 0.005\,\mathrm{V}$, $b = 10$, $\Lambda_{\mathrm{crit}} = 0.01\,\mathrm{V\,s}$, $\Lambda_{\max} = 0.015\,\mathrm{V\,s}$, $s_{\max} = 5\,\mathrm{nm}$, $\phi = 0.82\,\mathrm{eV}$, $A = 0.17 \times 10^{-18}\,\mathrm{m}^2$, $m_{\mathrm{eff}} = 0.99$. Calibrated `N4_PARAMS`: $V_{\mathrm{set}} = 0.06\,\mathrm{V}$, $b = 0.02$, $\Lambda_{\mathrm{crit}} = 0.01\,\mathrm{V\,s}$, `unipolar=False`, $V_{\mathrm{reset}} = r\,V_{\mathrm{set}} = 0.03\,\mathrm{V}$

(hold-band ratio $r = 0.5$), $\Lambda_{\max} = 1.5\,\Lambda_{\text{crit}}$. Growth time $\Lambda_{\text{crit}}/(|V|-V_{\text{set}}) \sim 0.1\,\text{s}$; dissolution $\Lambda_{\max}/(bV_{\text{reset}})$, washout tens of s. Step $2\,\text{ms}$ validated within 1% of $1\,\text{ms}$ ($5/10\,\text{ms}$ fail 5–12%).

Calibration (`fig4.py`): $3.6\,\text{V}$ sine, $50000 \times 1\,\text{ms}$, fresh net, two bands. Loss over $(I_{\max}, V_{\text{onset}}, A_+, A_-/A_+)$:

$$\text{loss} = \sum_k \left(\frac{m_k - t_k}{t_k}\right)^2$$

Paper $N = 2000$ targets: $I_{\max} = 3.24\,\text{mA}$, $V_{\text{onset}} = 1.92\,\text{V}$, $A_+ = 2.96\,\text{mA V}$, ratio 0.44. Fit loss 0.0085 (fitted onset $1.96 \pm 0.05\,\text{V}$ over 4 seeds). Fast dissolution ($b \geq 1$) gives mirror loops (ratio 1.00); fit needs $b \sim 0.005$–0.02. Caveat: Lamas Fig. 4 is itself a simulation, not measured ground truth.

Why it fails: frozen. Wide hold band $[V_{\text{reset}}, V_{\text{set}}]$ with ratio $r = 0.5$ parks most junctions where $d\Lambda/dt = 0$ (only $\sim 20/5953$ move); filaments saturate in $\sim 0.1\,\text{s}$ or persist $30\,\text{s}$. $\text{MC} = 0$, degree-2 ≤ 0.02. Narrow band $r = 0.9$ unfreezes (700–1700 move/sample, $\text{MC} \approx 1$) but still far from G2; full check N4-nb (E4 protocol at $r = 0.9$, see §7.1) not yet run (`n8_short.py` pilot only).

3.4 Volatile measured-data junction N10 — `agnw_rc.volatile`

Verdict: first positive result. Meets G1 on 16 pads; fragile to filament ruptures.

Built from Milano 2022 rate law (+ Diaz Schneider 2024 rupture, Milano 2023 consolidation). States per junction $x, y \in [0, 1]$: x volatile thin filament, y consolidated thick filament. Three rates act on x: growth k_p, decay k_d, and Joule melting k_{melt} (code name `k_P0` ; keep distinct from k_p).

$$\frac{dx}{dt} = k_p(V)\,\Theta(P_c - P)\,(1 - x) - [k_d(V) + k_{\text{melt}}(P)]\,x$$

$$k_p(V) = \min\left(k_{p0}\,e^{\eta_p s_V |V|},\, k_{\max}\right), \qquad k_{d,j}(V) = \frac{e^{-\eta_d s_V |V|}}{\tau_{s,j}}, \qquad k_{\text{melt}}(P) = k_{\text{melt},0}\,\Theta(P - P_c), \quad P = GV^2$$

Growth halts while $P > P_c$ (Joule melting). Zero-bias relaxation $\tau_{s,j} = 1/k_{d0,j}$, log-normal median τ_s, spread `tau_spread` (ln units), quenched seed $70000 +$ net seed. No hold band: relaxes at any sub-activation voltage. Non-polar ($|V|$ only).

$$\frac{dy}{dt} = k_c\,x^{n_c}(1 - y)\,\Theta(P - P_{c2}) - \frac{y}{\tau_l}$$

power-gated (formed junctions self-regulate ≈ 0.07–$0.1\,\text{V}$; power separates $\leq 4 \times 10^{-8}\,\text{W}$ at $2\,\text{V}$ vs median $2 \times 10^{-6}\,\text{W}$ at $3.5\,\text{V}$).

$$G = G_{\text{off}}(G_{\text{on}}/G_{\text{off}})^{\min(1,\,x+y)} \quad (\log); \qquad G = G_{\text{off}} + (G_{\text{on}} - G_{\text{off}})\min(1, x + y) \quad (\text{linear})$$

Exact step at fixed V (linear in own state):

$$x(t + \Delta t) = x_\infty + (x - x_\infty)e^{-(k_p + k_d + k_{\text{melt}})\Delta t}, \quad x_\infty = \frac{k_p}{k_p + k_d + k_{\text{melt}}}$$

$$y(t + \Delta t) = y_\infty + (y - y_\infty)e^{-(u_p + 1/\tau_l)\Delta t}, \quad u_p = k_c x^{n_c}\Theta(P - P_{c2})$$

unconditionally stable at any Δt. Noise options: additive $x \mathrel{+}= \sigma\sqrt{\Delta t}\ \mathcal{N}(0,1)$ clipped $[0,1]$; Poisson rupture $P(\text{hit}) = r_{\text{rup}}\,\Delta t$, $x \to x(1-f)$ with `rupture_frac` f (default 1 = full rupture).

Defaults (`VolatileParams`): $k_{p0} = 2.56 \times 10^{-6}\,\text{s}^{-1}$, $\eta_p = 34.9\,\text{V}^{-1}$, $k_{d0} = 64.9\,\text{s}^{-1}$ ($\tau_s = 1/64.9\,\text{s}$), $\eta_d = 5.59\,\text{V}^{-1}$, $v_{\text{scale}} = s_V = 1$, $k_{\max} = \infty$, $P_c = 10^{-3}\,\text{W}$, $k_{\text{melt},0} = 10^3\,\text{s}^{-1}$, $k_c = 0$, $n_c = 2$, $P_{c2} = 10^{-6}\,\text{W}$, $\tau_l = 10^3\,\text{s}$, `conductance="log"`, $\sigma = 0$, $r_{\text{rup}} = 0$ (`rupture_rate`). Project overrides (`config`): $v_{\text{scale}} = 5$, $k_{\max} = 100\,\text{s}^{-1}$ ($\geq 10\,\text{ms}$ switching; $2\,\text{ms}$ agrees with $1\,\text{ms}$ to 0.2% in current and 1.6% in states), $k_c = 0.1\,\text{s}^{-1}$ iff consolidation else 0. Milano numbers: $k_p = k_{d0}$ at $0.49\,\text{V}$; $k_p = 3\,\text{s}^{-1}$ at $0.4\,\text{V}$, $97\,\text{s}^{-1}$ at $0.5\,\text{V}$. $v_{\text{scale}} = 5$ maps to per-junction threshold $\approx 0.1\,\text{V}$; network onset $\approx 2.1\,\text{V}$ (two bands; 2.08–$2.2\,\text{V}$ across τ_s), $0.36\,\text{V}$ (pads). Best hold $\approx \tau_s/3$. Washout $\max(3\tau_s e^{2\,\text{spread}}, 3\,\text{s})$.

Why it works: many reproducible diverse fading variables (τ_s matched to input, no hold band, consistency 1.00); 16 electrodes expose them. N10-1 probes $\tau_s = 0.3$–$3\,\text{s}$: MC ≈ 2.0–2.4, degree-2 ≈ 3.4–4.1, NARMA ≈ 0.48–0.51, first G1 pass. N10-E pads: MC $= 3.48$, degree-2 $= 6.74$, NARMA $= 0.40$.

4. Interface

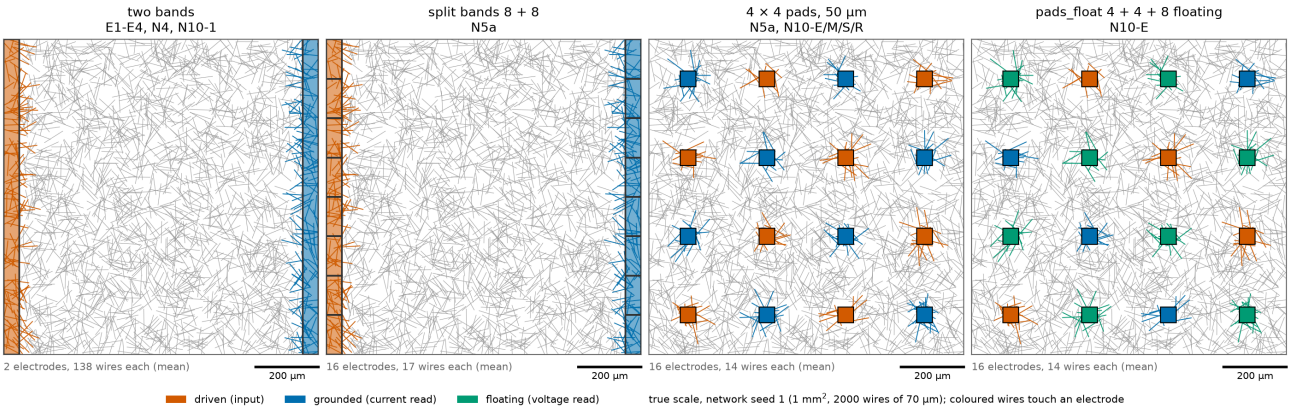


Figure 5. Electrode layouts at true scale on network seed 1; coloured wires touch an electrode.

4.1 Drive — `agnw_rc.drive`, `agnw_rc.multidrive`

Scalar $u_k \sim U(0,1)$, piecewise-constant voltage held `hold` steps ($1\,\text{step} = 1\,\text{ms}$; N4/N10 stride $2\,\text{ms}$):

$$V_k = V_{\text{lo}} + (V_{\text{hi}} - V_{\text{lo}})\,u_k, \quad V_{\text{lo/hi}} = V_c \mp V_{\text{width}}/2$$

Multi-electrode encodings ($x = 2u - 1$, `half` $= V_{\text{width}}/2$, weights $w_e \sim U(-1,1)$, seed $60000 +$ net seed):

- direct: $V_e = V_c + \text{half} \cdot x$ all e;
- weights: $V_e = V_c + \text{half} \cdot x \cdot w_e$;
- delays: $V_e(t) = V_c + \text{half} \cdot (2u(t - d_e) - 1)$, $d_e = 0, \ldots, n_{\text{in}} - 1$ (predrive fills history).

`simulate_multi` requires `spec.layout`; `hold % stride == 0`. Per step: solve circuit $\to dV \to$ junction update (`update_dynamics` / `meanfield_step` / `step_lambda` / `step_state`).

`drive.simulate` instead builds full $V(t)$ waveform and calls framework `run_simulation` unchanged, then recomputes probes/current from flags.

4.2 Electrode layouts — `agnw_rc.electrodes`

- `two_bands(band_frac=0.05)`: framework `assign_electrodes`, left 5% driven, right 5% grounded. Onset: binary $\approx 1.5\,\mathrm{V}$, N10 $\approx 2.1\,\mathrm{V}$.
- `split_bands(n_in=8, n_out=8)`: bands cut into equal-y segments `L0..R7`.
- `pad_grid(n_side=4, pad_size=50e-6)` ($50\,\mu\mathrm{m}$ pads): centers $((i+0.5)S/n, (j+0.5)S/n)$; default checkerboard in/gnd (8+8); `pads_float` 4 in + 4 gnd + 8 floating read pads; `pads_large` $100\,\mu\mathrm{m}$. N10 onsets: pads $0.36\,\mathrm{V}$, `pads_float` $0.73\,\mathrm{V}$, $100\,\mu\mathrm{m}$ pads $0.19\,\mathrm{V}$. Windows re-centered per layout from quasi-static onset sweep, width $0.45\times$ onset, positions $1.05\times$ and $1.25\times$ onset.

4.3 Observables and virtual nodes

- `current`: total I sampled at `virtual_offsets(hold, n_virtual)` = `unique(round(linspace(0, hold-1, min(n_virtual, hold))))` (10 default, 50 in E4/N10) — transients visible.
- `electrodes`: all 16 electrode currents (scaled $\times 10^3$, i.e. mA), tiled with virtual nodes + floating-pad voltages where present.
- `probes`: 32 internal node voltages, last step of block (needs micro-contacts; reference only).
- `junctions`: all states as oracle (stochastic packed bits; mean-field P float16; filament $\Lambda/\Lambda_{\mathrm{crit}}$ float32; volatile (x, y) float32).
- `on_fraction`: stochastic mean flags; mean-field mean P; filament $|\Lambda|/\Lambda_{\mathrm{crit}} \geq 1$; volatile $\min(1, x+y) \geq 0.5$.
- Headline readout (what a real chip measures): two bands $\rightarrow$ `current` with virtual nodes; pads $\rightarrow$ `electrodes`. `probes` and `junctions` show what information exists inside, not what is usable.
- Resolution floor `MEAS_RES = 1e-5` ($10\,\mathrm{nA}$ / $10\,\mu\mathrm{V}$); features below dropped. `v_in` recorded (mean driven V_e).

5. Learning, tasks, baselines, criteria

5.1 Ridge readout — `agnw_rc.readout`

Z-scored SVD ridge, unpenalized intercept. Keep features with $\sigma > 10^{-9}$, $Z = (X-\mu)/\sigma = USV^T$, weights $\mathbf{w}(\alpha) = \mathbf{V}\,\mathrm{diag}(s/(s^2+\alpha))\,\mathbf{U}^T\mathbf{y}_c$. `ALPHAS = logspace(-6, 4, 11)`. `fit_predict`: fit train, pick α per target (min MSE / max sign-accuracy on validation), refit train+val, predict test. Splits `SPLIT = (0.6, 0.2, 0.2)` time-ordered after washout, `start = 30`. Standardization uses training statistics only.

5.2 Tasks — `agnw_rc.tasks`

Row t state follows $u[t]$; targets use $u[0..t]$. Streams: `iid_stream` $U(0,1)$; `sinesquare_stream(period=8)` random whole periods of $\sin(2\pi k/8)$ vs ± 1 square, labels $0/1$. `delayed(u, k)` returns $u[t-k]$ (NaN if $t<k$); Legendre input $x = 2u - 1$, $P_1 = \sqrt{3}\,x$, $P_2 = \sqrt{5}\,(3x^2-1)/2$.

- Linear memory: targets $u[t-k]$, $k = 0, \ldots, 30$ (`MAX_LAG`).
- Degree-2 IPC: $P_2(x[t-k])$, $k < 11$, plus $P_1[t-k]P_1[t-l]$, $0 \le k < l \le 5$ (orthonormal under $U(-1,1)$).
- EMA integration: $z \leftarrow \lambda z + (1-\lambda)u_t$, $z_0 = 0.5$, $\lambda \in (0.5, 0.8, 0.9, 0.95, 0.98, 0.99, 0.995)$, $\tau = 1/(1-\lambda)$.
- NARMA-10: $y[t+1] = 0.3y[t] + 0.05y[t]\sum_{i=0}^{9} y[t-i] + 1.5s[t-9]s[t] + 0.1$, $s = 0.5u$, $y = 0$ before $t = 10$.
- Sine/square: per-sample classification from same stream.

5.3 Metrics — `agnw_rc.metrics` , `agnw_rc.evaluate`

$$\mathrm{NMSE} = \frac{\langle(\widehat{y}-y)^2\rangle}{\mathrm{Var}(y)}, \qquad C = \max(0,\, 1-\mathrm{NMSE}), \qquad \mathrm{acc} = \langle \mathbb{1}[\widehat{c}=c]\rangle$$

$$\mathrm{MC} = \sum_{k\ge 1,\, C_k > \mathrm{floor}} C_k, \qquad \mathrm{IPC}_2 = \sum_{C > \mathrm{floor}} C$$

floor from Gaussian-noise features (max over dims $(32, 1000)$, seeds $0, 1, 2$).

Consistency between two replicates a, b (same input, different noise and initial state): mean Pearson correlation over the F features with std $> 10^{-9}$ in both,

$$\mathrm{cons} = \frac{1}{F}\sum_{f=1}^{F}\frac{1}{T}\sum_{t=1}^{T} z^a_{tf}\, z^b_{tf}, \qquad z_{tf} = \frac{X_{tf} - \bar{X}_f}{\sigma_f},$$

NaN if no feature qualifies. Measured on state part (I/V_{in}, V_p/V_{in}; multi-electrode residual after `lstsq([V_electrodes, 1])`) and on its sample-to-sample increments (keys `*_state_inc` , from `np.diff`), so shared drift and V_{in} echo do not count. Why: raw consistency is inflated by two things that carry no computation: the input echo (current $\propto V_{\mathrm{in}}$ through a fixed conductance) and slow drift common to both replicates. Dividing by V_{in} (or regressing on the electrode voltages) removes the echo; differencing removes the drift. G0 uses the increment version. `evaluate.score_iid` / `score_sinesquare` return test + validation scores; observables scaled (current $\times 10^3$, i.e. mA).

5.4 Baselines — `agnw_rc.baselines`

Input-only u; delay line 32 taps (perfect linear memory, no nonlinearity); quadratic delay (8 taps + triu products); ESN $\mathbf{x} \leftarrow \tanh(\mathbf{W}\mathbf{x} + \mathbf{w}_{\mathrm{in}}(2u-1) + \mathbf{b})$, $\mathbf{W} \sim \mathcal{N}(0,1)$ scaled $\rho/\max|\mathrm{eig}|$, $\mathbf{w}_{\mathrm{in}} = \sigma_{\mathrm{in}} U(-1,1)$, $\mathbf{b} = 0.2U(-1,1)$, grid $\rho \in (0.5, 0.9, 1.1) \times \sigma_{\mathrm{in}} \in (0.1, 0.5, 1.5)$, hyper picked per task on validation; sizes $16/32/200$.

Reference scores depend on stream length:

baseline	2000 samples (E1–E3, M1, N2)	800 samples (E4, N10)
ESN-32: MC / IPC2 / NARMA	13.8 / 14.4 / 0.24	13.5 / 14.2 / 0.23
ESN-200: MC + IPC2	51.5	50.1
delay-32: MC / IPC2 / NARMA	30 / 0 / 0.16	30 / 0 / 0.15
input-only NARMA	0.80	0.84

ESN-16 (loose G2 reference) MC + IPC2 $= 15.4$.

5.5 Criteria — `agnw_rc.criteria`

Fixed 2026-09-27, before E4 and N10 were run. Operating point (`hold`, `v_center`) = max mean `val_MC + val_IPC2` over iid runs. Choosing on validation, not test, was a review fix (REVIEW M5 / P0.2): earlier tables picked the best setting on test scores, which flatters the network; ESNs were already tuned on validation. Verdicts use test scores only, 3 nets $\times 2$ replicates, 2-sd margin:

- **G0** reproducible: mean increment consistency ≥ 0.8.
- **G1** computes beyond input: G0 and mean-2sd MC ≥ 1 (with ≥ 2 lags above floor) and IPC2 ≥ 1 and mean$+2$sd NARMA ≤ 0.70 (input-only ≈ 0.80–0.84).
- **G2** useful: G1 and mean MC+IPC2 $\geq 50\%$ of ESN same size and mean sine/square ≥ 0.95. Multi-electrode: **strict** (headline; ESN as large as feature count incl. virtual nodes) and **loose** (one ESN unit per physical electrode; ESN-16 total $15.4 \rightarrow \geq 7.7$). Strict uses the closest ESN size $\geq$ the feature count, else the largest computed (ESN-200, MC + IPC2 $= 50.1$, threshold 25; 16 electrodes $\times$ 50 virtual nodes = 800 features). Why two: strict compares equal readout sizes, the fair software comparison. Loose compares per physical contact, the hardware question (what 16 wires into a chip give vs 16 software units).
- Drift: $(\langle \mathrm{ON} \rangle_{\text{last } 20\%} - \langle \mathrm{ON} \rangle_{\text{first } 20\%}) / \langle \mathrm{ON} \rangle$, $|\mathrm{drift}| > 2\%$ not stationary.
- Washout $\geq 3\times$ slowest τ precedes scoring (`model_washout_steps`: binary/mean-field $\max(30000, 3/p_{\text{relax}})$; filament $3\Lambda_{\max}/(bV_{\text{reset}})/\Delta t$; volatile $\max(3\tau_s e^{2\,\text{spread}}, 3\,\text{s})/\Delta t$).

6. Code reference (every file)

6.1 `agnw_rc/` — reservoir layer

file	does	key API
`config.py`	canonical working model + `RunSpec` + all grids (E1–E4, M1, N2, N5a, N7a/b, N4, N10-1/E/M/S/R); `VOLATILE_V_SCALE=5`, `VOLATILE_K_MAX=100`, `VOLATILE_K_C=0.1`	`network_params()`, `make_network(seed)`, `RunSpec`, `model_washout_steps()`, `volatile_params()`, `filament_params()`, `*_specs()`

file	does	key API
`drive.py`	two-electrode pipeline through `run_simulation` unchanged; streams, waveform, snapshots, probe pick `rng(40000+seed)`, save/load `.npz` + `framework_commit`	`input_stream()`, `to_voltage()`, `virtual_offsets()`, `simulate()`, `raw_path()`, `save_raw/load_raw`
`multidrive.py`	multi-electrode time loop, encodings `weights/direct/delays`, per-model step, readouts incl. `v_float`, `probe_v`, junction states, `on_fraction`	`input_weights()`, `electrode_voltages()`, `simulate_multi()`
`multicircuit.py`	multi-terminal Kirchhoff on framework graph; reuse/direct/pcg solves; stats	`MultiTerminalCircuit.solve()`, `junction_voltage_drops()`, `electrode_currents()`
`electrodes.py`	layouts + contact geometry (`_band_contact_t`, Liang–Barsky clip, pad crossings at $t = 0.5(u_0 + u_1)L$)	`two_bands()`, `split_bands()`, `pad_grid()`, `make_layout()`, `LAYOUTS`
`meanfield.py`	N1/N2 expectation dynamics + `MeanFieldSolver` PCG	`meanfield_step()`, `simulate_meanfield()` (draws the N2 $p_{\mathrm{relax},\,j}$ spread), `MeanFieldSolver`
`filament.py`	N4 rate law + Simmons + Euler–Maruyama noise + conductance	`FilamentParams`, `lambda_rate()`, `step_lambda()`, `gap()`, `tunnel_conductance()`, `conductance()`
`fig4.py`	Fig. 4 sine protocol + framework envelope metrics + calibration loss	`sine_protocol()`, `metrics_fn()`, `paper_targets()`, `loss()`, `LOSS_KEYS`
`volatile.py`	N10 x/y exact steps + log/linear G + additive/rupture noise + quenched τ_s	`VolatileParams`, `junction_kd0()` (quenched $1/\tau_{s,\,j}$), `rates()`, `step_state()`, `conductance()`

file	does	key API
`tasks.py`	streams, delays, Legendre, memory/IPC/EMA/NARMA/sinesquare, splits	`iid_stream()`, `memory_targets()`, `ipc2_targets()`, `ema_targets()`, `narma10_target()`, `split_indices()`
`readout.py`	SVD ridge, per-target alpha	`Ridge.fit_svd()`, `fit_predict()`, `ALPHAS`
`metrics.py`	NMSE/capacity/floor-sum/accuracy/consistency	`nmse()`, `capacity()`, `thresholded_sum()`, `consistency()`
`baselines.py`	no-nanowire references + ESN grid	`delay_line()`, `quadratic_delay_line()`, `esn_states()`, `noise_features()`
`evaluate.py`	observables (current $\times 10^3$), scoring, state/inc consistency, noise floor, `MEAS_RES`	`observables()`, `score_iid()`, `score_sinesquare()`, `noise_floor()`, `iid_baselines()`, `state_observables()`, `evaluate_consistency()`
`criteria.py`	G0/G1/G2 + operating point + drift	`select_operating_point()`, `evaluate()`, `drift()`

6.2 `scripts/` — pilots, runs, reports, checks

script	does
`pilot.py`, `meanfield_pilot.py`, `multi_pilot.py`	operating-point pilots → `results/*.npz`
`run_sims.py`	parallel resumable runs (`e1 e2 e3 m1 n2 n7a n7b n5a n4_1 e4 n10_1 n10m`; `--holds/--reverse/--limit` split across machines; `--blocks` for n10m) → `results/raw/*.npz`
`e4_report.py`, `n10_report.py`, `n10m_report.py`	go/no-go reports → `results/*_criteria.json`, figures
`n10e_pilot.py`	`sweep/run/report/full/full_report` for N10-E pads

script	does
`n10s_sensitivity.py`, `n10r_noise.py`	N10-S $v_{\text{scale}} \times k_{\max}$; N10-R rupture sweep/run/report/twins (each low-rate run vs its rupture-free twin)
`n10l_size.py`	N10-L: larger networks (`RunSpec.side` , layout `pads_scaled`), sweep/run (`--share i/n` splits runs between machines)/report
`n10l_seeds.py`	N10-LS: networks 4-10 per size, combined noise, 4 mm; run (atomic `.running` claims, safe on a shared cluster filesystem; `--only SIZE:RATE[:SIGMA]`)/report (pooled G1, networks passing, spread, ρ) -> `results/n10l_seeds.json` , fig31; cluster job `cluster/run_seeds.sh`
`n10r_retrain.py`, `n10b_bias.py`, `n10v_vscale.py`	N10-OR causal retraining of the readout (no simulations); N10-B bias + small input (run/report); N10-V `v_scale` 40 (sweep/check/run/csweep/crun/report)
`n10e_readout_noise.py`, `n10k_consolidation.py`	N10-RN: measured network-level noise on the electrode currents of N10-E (no simulations); N10-K: forming + consolidated backbone, pilot/run/report
`n10c_data.py`, `n10c_sim.py`	N10-C: measured statistics of the Milano 2025 open data (`data/milano2025/`); their stationary protocol on N10 two-band networks (sweep/run/report; `.running` markers prevent duplicate runs)
`evaluate_all.py`	all scores $\rightarrow$ `results/metrics.json` , `summary.csv`
`n4_stride_check.py`, `n4_substep_check.py`, `n10_kmax_check.py`, `n10_stride_check.py`	step validation ($2\,\text{ms}$ OK, 5–$10\,\text{ms}$ fail; $k_{\max} > 100$ needs $1\,\text{ms}$)
`n8_short.py`	N4 hold-band vs forgetting scan $\rightarrow$ `results/n8_short.json`
`switching_rates.py`, `noise_averaging.py`, `leaky_bank_reference.py`, `effective_dimension.py`	mechanism rates, $R = 1$–8 averaging, ideal leaky bank, dimensionality
`n4_calibrate.py`, `n4_model_figures.py`, `n4_bipolar_check.py`	Fig. 4 fit, model plots, polarity check

script	does
`n4_refine_validate.sh`	chains `n4_calibrate.py refine` then `validate` on the best point (runs from the project root)
`n10_phenomenology.py`, `n10_power_gate.py`, `n10e_channels.py`, `esn16_baseline.py`	M1–M8 checks, consolidation gate, per-channel info, ESN-16 reference
`make_figures.py`	all result figures (`--only` , `--mode paper|presentation`)
`manual_figures.py`	the five figures of this manual → `figures/manual/` (PNG + SVG; `--only` , `--mode paper|presentation`); physics panels drawn from `agnw_rc` model code, layouts from network seed 1
`review_checks.py`, `md_math_to_latex.py`	review evidence table; LaTeX math fixer (dry run default)

6.3 `tests/` , `docs/` , rest

`test_core.py` framework contract (preset values, solver, commit), ridge, capacity/NMSE, consistency, delay-line and quadratic baselines; `test_drive.py` waveform/snapshot alignment; `test_phase0.py` washout, virtual nodes, stride, drift, operating point, G0–G2; `test_meanfield.py`, `test_filament.py` , `test_filament_network.py` , `test_volatile.py` , `test_multicircuit.py`, `test_multidrive.py` (model math, solver match, encodings).

`docs/PLAN.md` design; `docs/RESULTS.md` full results (§28 reproduce); `docs/NEXT_STEPS.md` status/log (update every change); `docs/SWITCHING_MECHANISMS.md` junction physics + framework audit; `docs/N4_DESIGN.md` , `docs/N10_DESIGN.md` model derivations; `docs/REVIEW_2026-09-27.md` bug corrections; `docs/LITERATURE_NOTES.md` , `docs/MEASUREMENT_REQUEST.md` (rupture rate = priority 1); `docs/report_2026-09-28/` , `docs/report_2026-09-29/` (md/html/pdf), `docs/report_2026-10-06/` (md/pdf) summary reports; `docs/MANUAL.pdf` built from this file by `docs/md_to_pdf.sh MANUAL.md` ; `docs/md_to_html.sh <report.md>` builds a self-contained HTML report (pandoc + headless Chromium). `legacy/` superseded prototype — do not cite its capacities.

7. Experiments, outputs, reproduce

7.1 Experiment glossary

The wire network never changes; experiments vary junction model and interface. Section numbers refer to `docs/RESULTS.md` .

name	question	junction	layout / readout	outcome	RESULTS §
E1–E3	baseline sweep: hold, voltage, p_{relax}	binary	two bands; current, probes	input-only level	1–5
N1 / M1	is noise the problem?	mean-field	two bands	reproducible, nonlinear, MC ≈ 1.8	7
N2	heterogeneous forgetting	mean-field, spread p_{relax}	two bands	$\approx 2\times$ memory, still noise-limited	8
N7a	average noise away?	binary, 8 replicates	two bands	no rescue	9
N7b	is M1 gain just higher voltage?	binary at M1 windows	two bands	no; gain is noise removal	10
N5a	multi-electrode input	binary	split bands, pads; weights	no gain	11
N4-1	physical graded junction	N4 filament	two bands	frozen, capacity 0	12
E4	inputs at network time scale (0.3–3 s)	binary, N4	two bands; 50 virtual nodes	G1 fails both	13
N8 (short)	hold band vs forgetting	N4 variants	two bands	hold band freezes; $r = 0.9$ weak	14
N4-nb	E4 protocol at $r = 0.9$	N4 narrow band	two bands	**open**	—
N10-1	measured-data volatile junction	N10	two bands; 32 probes	G1 pass, G2 no	15
N10-E	N10 read via 16 electrodes	N10	pads; electrode currents	G1; G2 loose yes, strict no	16–17

name	question	junction	layout / readout	outcome	RESULTS §
N10-M	memory limit, noise tolerance	N10 + spread/delays/ σ	pads	memory ≈ 4 inputs; $\sigma \geq 0.1$ breaks G1	18
N10-S	sensitivity to chosen settings	N10, v_{scale}, $k_{\max}$	pads	robust	19
N10-R	filament ruptures + mitigation	N10 + rupture	pads	tolerance $\approx 10^{-4}\,\mathrm{s}^{-1}$ per junction; above it G1 breaks	20
N10-C	calibrate N10 on Milano 2025 data	N10 + rupture / σ	two bands, 1–2.83 mm, sampled every 0.6 s	$r^{*} \approx 0.005$–$0.03\,\mathrm{s}^{-1}$ (50–$300\times$ tolerance)	21
N10-RN	measured noise on the readout	N10 (existing runs)	pads	harmless at 1x; G1 to 3x	22
N10-K	consolidated backbone vs ruptures	N10 + y, forming	pads	collapses gone, capacity halved; no G1	23
N10-OR	online retraining of the readout	N10-R runs	pads	no help	24
N10-B	bias + small input	N10 (+ ruptures)	pads	best noise-free (MC 5.8); most fragile	25
N10-V	`v_scale` 40 (matched)	N10 (+ ruptures)	pads; two bands	unchanged; calibration unchanged	26
N10-L	rupture tolerance vs network size	N10 + ruptures	pads_scaled, 1.41 / 2 / 2.83 mm	G1 at 10^{-2} /s for 2.83 mm (real-junction range); size buys	27

name	question	junction	layout / readout	outcome	RESULTS §
				reliability, ~half capacity lost	
N10-LS	N10-L with 10 networks per size	N10 + ruptures (+ σ, readout noise)	pads_scaled, 1 / 2 / 2.83 mm (10 networks), 4 mm (3)	2.83 mm: 9/10 networks pass G1, pooled G1 fails (NARMA margin); 4 mm passes	27 (addendum)
N3 / N9	input masks / dense virtual nodes	any	interface	N9 used in E4/N10; N3 open	—

7.2 Grids — `agnw_rc.config.*_specs()`

`RunSpec` fields: `stream, hold, v_center, network_seed, replicate, v_width=1.0, p_relax=1e-4, n_samples=0, model="stochastic", p_relax_max=0.0, layout="", encoding="weights", sigma=0.0, washout=0, n_virtual=0, stride=1, tau_s=0.0, tau_spread=0.0, consolidation=False, v_scale=0.0, k_max=0.0, rupture=0.0, k_c=0.0, tau_l=0.0, forming=0.0, forming_v=0.0, p_c2=0.0, side=0.0` (k_c … p_c2: N10-K; side: N10-L network side in mm; names unchanged when 0). Properties: `washout_steps = washout or 10000`, `n_virtual_eff = n_virtual or 10`, `length = n_samples or 2000 (iid) / 1200 (sinesquare)`, `v_lo`, `v_hi` (§4.1).

- E1: holds $(10, 30, 100)\,\mathrm{ms}$, centers $(1.25, 1.75, 2.25)\,\mathrm{V}$.
- E2: $p_{\mathrm{relax}} \in (0, 10^{-4}, 10^{-3}, 10^{-2})$.
- E3: holds $(300, 1000)\,\mathrm{ms}$, 800 samples.
- M1/N2: mean-field, centers $(3.5, 4.5, 5.5)\,\mathrm{V}$; N2 $p_{\mathrm{relax,\,max}} \in (10^{-3}, 10^{-2}, 10^{-1})$.
- N7a: 8 replicates at hold $30\,\mathrm{ms}$, 1.25–$2.25\,\mathrm{V}$ (binary).
- N7b: binary at M1 windows $(3.5, 4.5, 5.5)\,\mathrm{V}$, holds $(10, 30)\,\mathrm{ms}$.
- N5a: `split_bands` / `pads`, weights encoding, holds $(10, 30)\,\mathrm{ms}$.
- N4-1: filament, two bands, centers $(1.75, 2.25, 2.75)\,\mathrm{V}$.
- E4 decisive: holds $(300, 1000, 3000)\,\mathrm{ms}$, 800 samples, 50 virtual nodes, stride 2.
- N10-1: $\tau_s \in (0.03, 0.3, 3.0)\,\mathrm{s}$, holds $(\tau_s/3, \tau_s)$, spreads $(0, 1)$, centers $(2.25, 2.75)\,\mathrm{V}$, stride 2.
- N10-E/M/S/R: 4×4 pads, windows (center, width) $(0.38, 0.16)$ and $(0.45, 0.16)\,\mathrm{V}$, τ_s 1–3 s, spreads, delays, $\sigma = 0.03/0.1/0.3$, $v_{\mathrm{scale}} \in$ 3–8, $k_{\max} \in$ 30–300 s^{-1}, ruptures 10^{-5}–3 s^{-1}.

- N10-C (`scripts/n10c_sim.py`, own grid): two bands, $nL^2 = 9.8$ (sides $1, 1.41, 2, 2.83\,\mathrm{mm}$) and 5.8 ($1\,\mathrm{mm}$), bias $1.5\times$ onset, $300\,\mathrm{s}$ sampled every $0.6\,\mathrm{s}$; ruptures 0–$1\,\mathrm{s}^{-1}$, σ 0.01–0.1 (above $1\,\mathrm{mm}$ only $r = 0, 10^{-4}, 10^{-3}$, $\sigma = 0.01$). Raw output `results/raw/n10c/*.npz` (keys `t`, `G`, `bias_V`, `onset_V`, `n_junctions`, `x_final`).

7.3 Outputs

Raw runs `results/raw/<RunSpec.name>.npz` (`drive.raw_path`; written atomically via `.tmp.npz`). The name encodes the spec, e.g. `iid_h1000_v0.38_w0.16_pr1e-04_n1_r0_s800_pads_weights_vol_ts3_sp1_wo67000_nv50_st2`: stream, `h` hold, `v` center, `w` width, `pr` p_{relax}, `n` net seed, `r` replicate, then optional `s` samples, `mf`, `prmax`, `<layout>_<encoding>`, `fil[_sig]`, `vol_ts<tau_s>_sp<spread>[_con][_vs][_km][_rup][_sig]`, `wo` washout, `nv` virtual nodes, `st` stride.

Keys (all runs): `u`, `labels`, `v_in`, `current`, `virtual_offsets`, `probe_v`, `probe_nodes`, `n_junctions`, `on_fraction`, `solver_stats`, `seconds`, `framework_commit`, and `spec_<field>` for every `RunSpec` field. Junction state by model: `flags_packed` (binary, `np.packbits`; unpack with `n_junctions`), `junction_p` (mean-field), `junction_lambda` ($\Lambda/\Lambda_{\mathrm{crit}}$), `junction_x`, `junction_y` (volatile). Junction states are stored only with `simulate_multi(..., save_states=True)` (default `config.SAVE_JUNCTION_STATES = False` since 2026-10-08). Runs before that date were slimmed locally (`scripts/slim_raw.py`: `junction_x/y`, `junction_p` removed, key `junction_states_archived` added); the full files are on the external drive TOSHIBA_ntfs, `AgNW_Reservoir_Computing_archive/raw_simulation_runs_with_junction_states_2026-10-08/raw/`. Without states, `agnw_rc.evaluate` simply has no "junctions" readout. Two-electrode only: `i_recomputed`, `i_simulated`, `i_total_scored`, `i_total_predrive`. Multi-electrode only: `v_electrodes`, `stride`, `n_electrodes`, `electrode_roles`, `electrode_n_wires`, `v_float` (if floating pads); `current` holds all electrode currents.

Scores: `results/metrics.json` (per run), `results/summary.csv` (mean/std over nets and replicates), `results/*_criteria.json` (operating point + G0/G1/G2 per readout), other `results/*.json` from checks.

7.4 Reproduce

```
.venv/bin/python scripts/pilot.py
.venv/bin/python scripts/run_sims.py e1 --procs 14
.venv/bin/python scripts/run_sims.py e4 --model stochastic --procs 12
.venv/bin/python scripts/run_sims.py n10_1 --procs 16
.venv/bin/python scripts/n10_report.py
.venv/bin/python scripts/n10e_pilot.py sweep
.venv/bin/python scripts/n10e_pilot.py full --procs 18
.venv/bin/python scripts/run_sims.py n10m --procs 18
.venv/bin/python scripts/evaluate_all.py
.venv/bin/python scripts/make_figures.py
.venv/bin/python -m pytest -q   # 80 tests, ~7 min
```

Full list with outputs: `docs/RESULTS.md` §28.

Second machine `ssh ultron`: rsync code excluding `results/raw`, `metrics.json`; launch via `/tmp` script with `setsid nohup`; split `--reverse/--limit`; pull with `rsync -a --ignore-existing --exclude '*.tmp.npz'`. Its project copy is still in the old location (`~/Desktop/...`, as of 2026-09-30): re-rsync the current code before using it; ask Juan for availability.

8. Limits and next steps

Bottom line: N10 computes beyond its input (best: bias + small input, MC 5.8, above an ESN-16) but realistic filament ruptures erase the gain. Real networks imply rupture rates 50–300$\times$ the tolerance (N10-C), and no remedy tried so far restores G1 at that level.

- Memory capped at ≈ 4 inputs for any τ_s; delays add only interface memory ($\mathrm{MC} = 7.6 =$ delay line $7 + 0.6$).
- Junction noise: $\approx 1\%$ of filament range tolerated, $\approx 4\%$ ($\sigma \geq 0.1$) breaks G0/G1.
- Ruptures: tolerance $r_{\mathrm{rup}} \approx 10^{-4}\,\mathrm{s}^{-1}$ per junction; single runs collapse from 10^{-4}, every run loses about half from 10^{-3} (consistency down to 0.36 at $3\,\mathrm{s}^{-1}$); no mitigation (bigger pads, longer holds, double presentation) restores G1.
- Measured-size fluctuations on the readout are harmless near onset (N10-RN, §22): the damage is ruptures rerouting current inside a small network.
- Remedies that fail at realistic noise (RESULTS §23–26): consolidated backbone (collapses gone, capacity halved), online retraining of the readout (never better than static), bias + small input (best noise-free, most fragile), v_{scale} matched to Milano (unchanged).

Decisive unknown: real per-junction rupture rate — N10-C (RESULTS §21) maps the Milano 2025 open data (Zenodo `10.5281/zenodo.15050217`) onto it: ≈ 0.005–$0.03\,\mathrm{s}^{-1}$, 50–$300\times$ above the tolerance. N10 junctions switch at $\sim 10\times$ their voltage, but matching it (v_{scale} 40, N10-V) changes neither the reservoir nor this estimate; small-network / single-junction measurements (`docs/MEASUREMENT_REQUEST.md`). Network size works, mostly (N10-L / N10-LS, §27): at $10^{-2}\,\mathrm{s}^{-1}$, inside the real-junction range, 9 of 10 networks of 2.83 mm meet G1 (pooled G1 fails on the NARMA-10 margin; 4 mm passes, 3 networks); noise still costs about half the capacity at every size (size reduces the spread between networks). Next: size + bias / small input, more electrodes on larger networks, N10-K with re-tuned windows / partial backbones, N4-nb, Mackey-Glass / spoken digits. Current status: `docs/NEXT_STEPS.md`.

9. Extending: new junction model or experiment

Rule: the framework is only imported; new physics lives here, and a special case of it must reproduce the framework exactly (test it).

New junction model (pattern of N4 / N10):

1. `agnw_rc/<model>.py`: a frozen params dataclass and pure functions `rates` / `step` / `conductance` on arrays of junction drops (vectorized, exact or validated step at `dt = stride` ms). Unit tests on the model math.
2. `config.py`: `<model>_params(spec)` building the params from a `RunSpec` (framework G_{on}, G_{off}; `dt = spec.stride * DT`); washout rule in `model_washout_steps` ($\geq 3\times$ slowest time).
3. `RunSpec`: new fields with defaults that leave existing run names unchanged; a name suffix in `RunSpec.name` (otherwise runs overwrite each other in `results/raw/`).
4. `multidrive.simulate_multi`: dispatch on `spec.model` (initial state, per-step update, oracle state `junction_*` key, `on_fraction` rule). Add the key to `evaluate.observables`.
5. Tests: a limit where the model equals an existing one (e.g. framework binary or N10 at default settings) gives the same currents to round-off; step-size check script like `n10_stride_check.py`.

New experiment:

1. `config.<name>_specs()` returning the grid of `RunSpec`s (3 network seeds $\times$ 2 replicates + sinesquare runs if criteria are needed); voltage windows from an onset sweep for each new layout or model.
2. Add the name to `run_sims.py` choices (or a dedicated script with `sweep/run/report`, like `n10e_pilot.py`).
3. Report: `criteria.select_operating_point` + `criteria.evaluate` $\rightarrow$ `results/<name>_criteria.json`.
4. Document: status + log in `docs/NEXT_STEPS.md`, a section in `docs/RESULTS.md`, a row in §7.1 here, a box in Figure 2 (`scripts/manual_figures.py`, `fig2_taxonomy`).

10. Runtimes and troubleshooting

Typical cost on the main machine (one process per run, `OMP_NUM_THREADS=1`):

job	time
test suite (80 tests)	~7 min (with other jobs running)
N10 run, 800 samples, 1 s holds	~2 h
N10 run, 800 samples, 3 s holds	~4–6 h (ultron ~1.5$\times$ slower)
`manual_figures.py` (all five)	< 1 min

symptom	cause / fix
`test_framework_commit` fails	wrong framework checkout; check out commit `f6b6951` in `../AgNW_FDT_Memristor_Framework`

symptom	cause / fix
`ModuleNotFoundError: agnw_sim` or paths to an old location	`.venv` holds absolute paths: after moving the project, delete it and reinstall (`RESUME_FROM_HERE.md` §3)
machine load far above the number of processes	BLAS threads not limited: run scripts as given (they set `OMP_NUM_THREADS=1`) or export it
run restarts from zero after a crash	expected for that run only: finished runs are skipped (resumable), `*.tmp.npz` are partial writes
runs silently overwrite each other	new `RunSpec` field without a name suffix (§9)
`pkill -f` kills your own shell/ssh	never use a pattern contained in the running command line

11. References

- Lamas, Medina, Arlego, Matera (2026), *Modelado Estocástico de Redes de Nanohilos de Plata para Computación Neuromórfica* (JAIIO): binary junction model and Fig. 4 (framework preset; PDF in the framework's `biblio/`).
- Hochstetter et al. (2021), *Nat. Commun.* 12, 4008, doi:10.1038/s41467-021-24260-z, and Author Correction (2024), doi:10.1038/s41467-024-54027-1: N4 filament model.
- Milano et al. (2022), *Nat. Mater.* 21, 195: N10 rate law (parameters via Cipollini et al. 2024, arXiv:2411.19353).
- Milano et al. (2023), *Nat. Commun.* 14, 5723: consolidation (long-term memory).
- Milano, Michieletti, Pilati, Ricciardi, Miranda (2025), *Nat. Commun.*, doi:10.1038/s41467-025-58741-2: open dataset for N10-C (Zenodo `10.5281/zenodo.15050217`).
- Diaz Schneider, Quinteros, Levy, Martínez (2024), arXiv:2409.01318: power-driven filament rupture.
- Diaz-Alvarez et al. (2019), *Sci. Rep.* 9, 14920: random relaxation times and abrupt drops.

Notes on each, and what was and was not verified: `docs/LITERATURE_NOTES.md`, `docs/N4_DESIGN.md`, `docs/N10_DESIGN.md`.